

Project 1: Multi-Account AWS Governance Setup using AWS Organizations and SCP

1. Objective

The primary objective of this project is to implement centralized governance across multiple AWS accounts by:

- Enforcing security policies
- Restricting unauthorized actions
- Controlling cost usage
- Maintaining compliance across environments

This is achieved using AWS Organizations and Service Control Policies (SCPs), ensuring that all accounts follow predefined rules regardless of individual user permissions.

2. AWS Organization Setup

The first step in implementation was setting up AWS Organizations.

- AWS Organization was created with **All Features Enabled**
- Organizational Units (OUs) were created:
 - Dev OU
 - Test OU
 - Prod OU
- Member accounts were added and organized under respective OUs

This structure allows centralized management and policy enforcement across all accounts.

Figure 1: Organizational Units Created

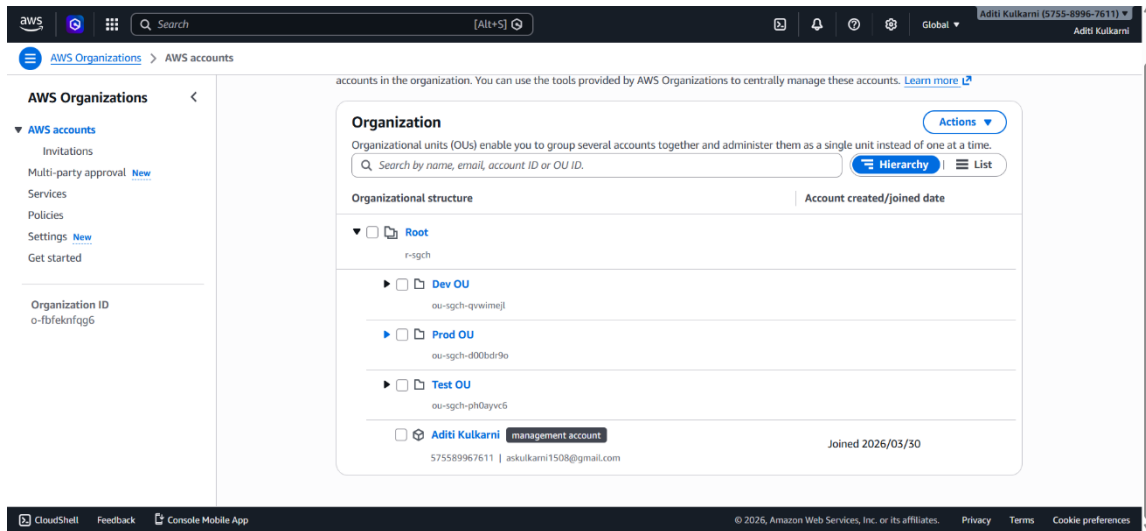
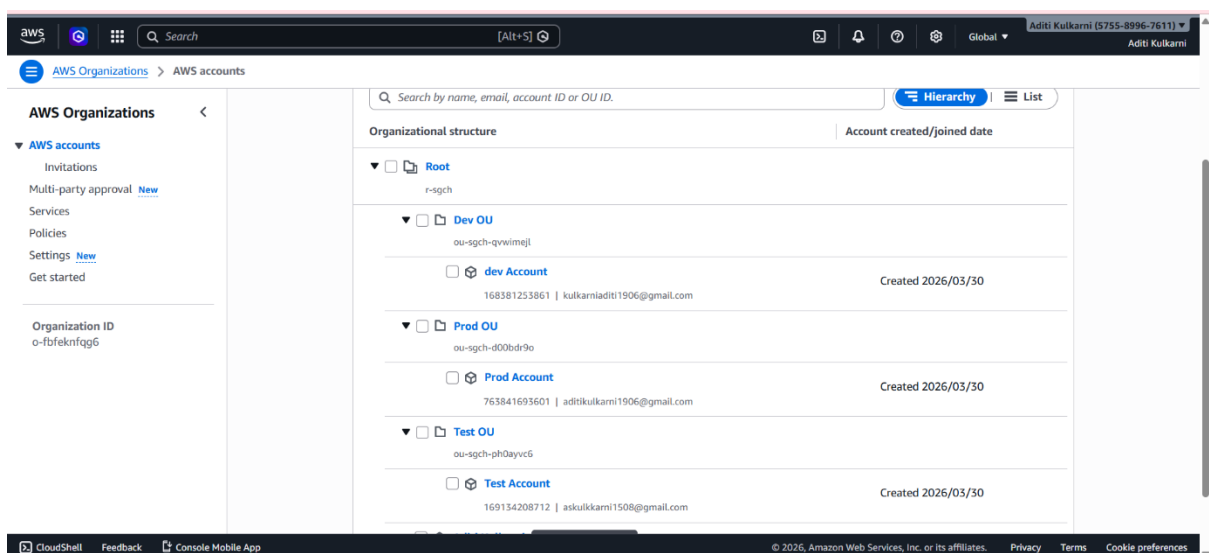


Figure 2: Accounts under Organizational Units



3. Service Control Policies (SCP) Implementation

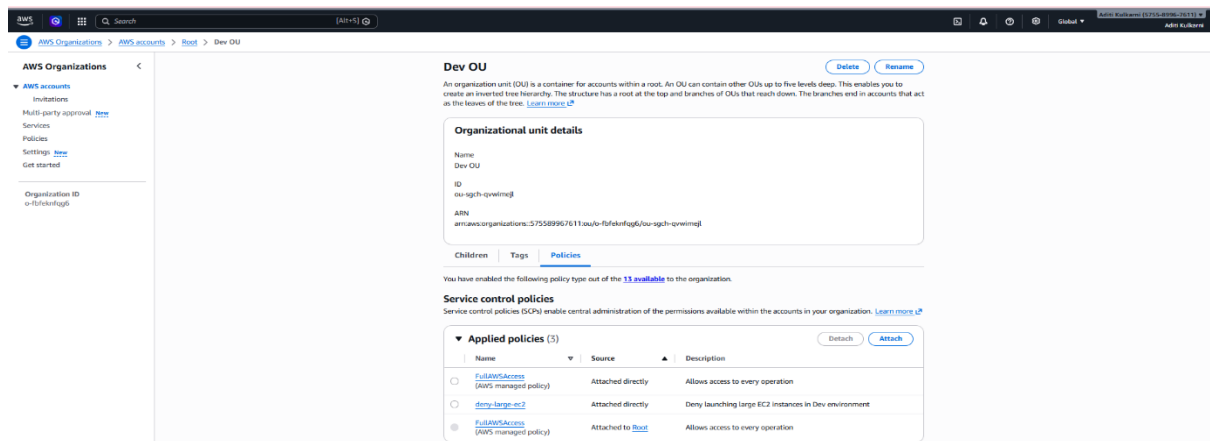
Service Control Policies were created to enforce governance rules across accounts. These policies act as permission boundaries and override IAM permissions if needed.

Policy 1: Restrict Large EC2 Instances in Dev

This policy prevents developers from launching expensive EC2 instances in the Dev account. Only small instance types are allowed.

👉 Benefit: Prevents high cloud costs during development

Figure 3: SCP Attached to Dev OU

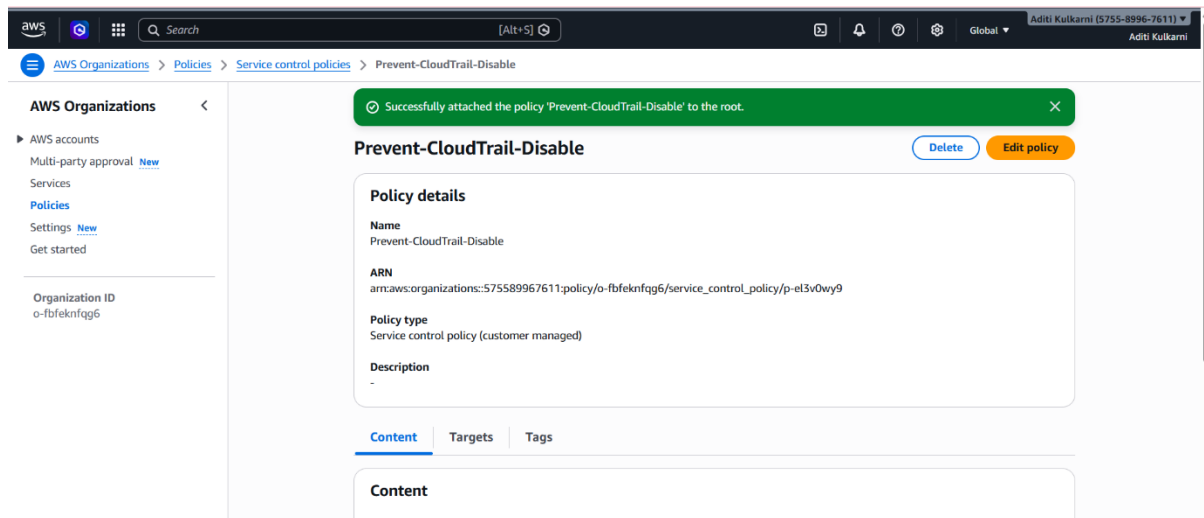


Policy 2: Prevent CloudTrail Disabling

This policy ensures that CloudTrail logging cannot be stopped or deleted in any account.

👉 Benefit: Ensures continuous monitoring and auditing

Figure 4: CloudTrail Protection SCP

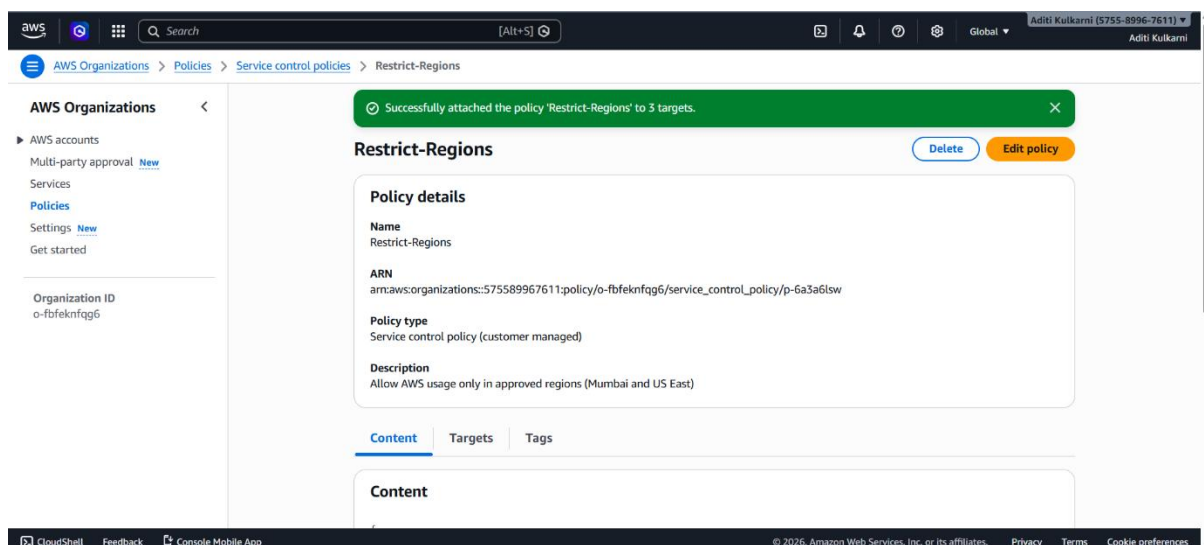


Policy 3: Restrict Region Usage

This policy allows resource creation only in approved regions (e.g., Mumbai and N. Virginia) and blocks all other regions.

👉 Benefit: Prevents unauthorized deployments and improves compliance

Figure 5: Region Restriction SCP



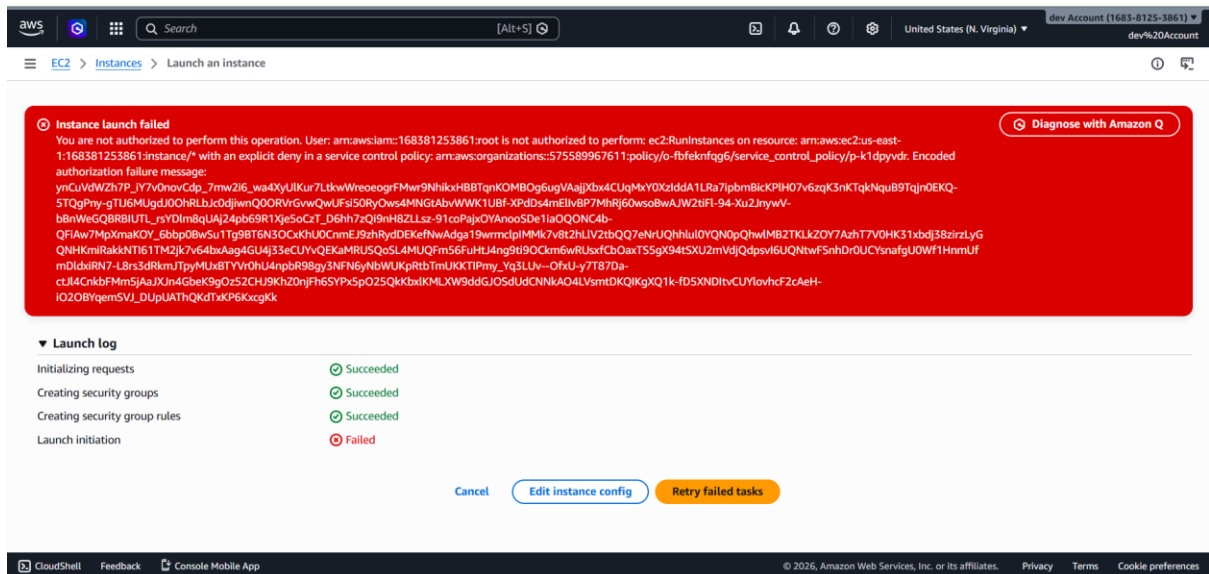
4. Validation and Testing

After implementing SCPs, validation was performed to ensure policies were working correctly.

Test 1: EC2 Instance Restriction

- Attempted to launch a large EC2 instance in Dev account
- Operation was denied

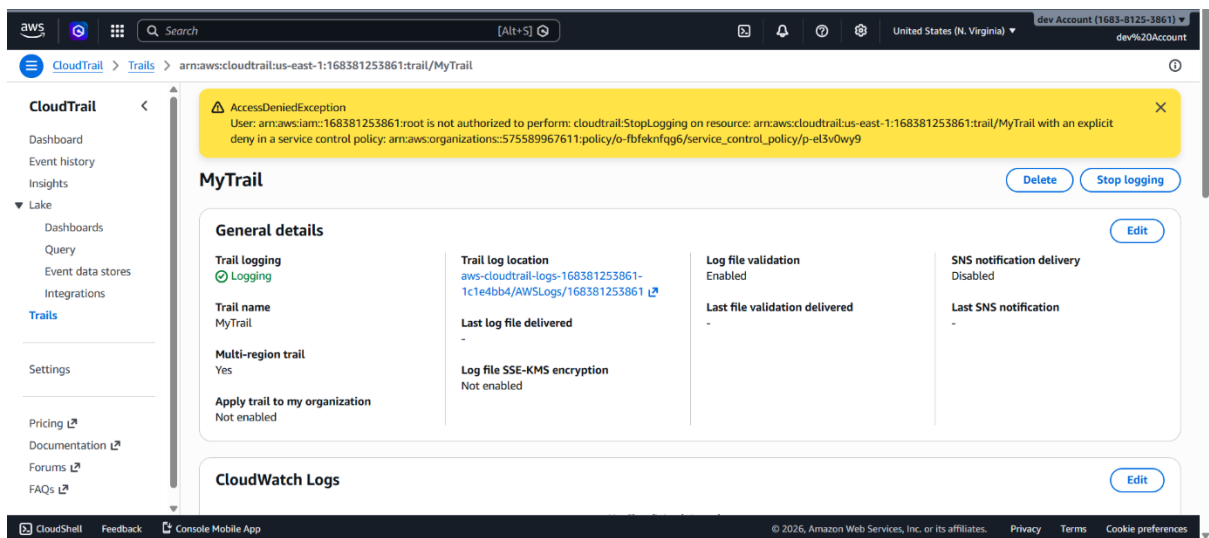
Figure 6: EC2 Launch Denied



Test 2: CloudTrail Protection

- Attempted to stop CloudTrail logging
- Operation was denied

Figure 7: CloudTrail Denied



Test 3: Region Restriction (Key Validation)

- Switched to a restricted region (e.g., Seoul)
- Attempted to create a key pair

The system returned the following error:

“Explicit deny in a service control policy”

This confirms that SCP is successfully enforcing restrictions.

Figure 8: SCP Explicit Deny Error

